

SINGLE SOURCE

SHORTEST PATH

(SSSP)

Input: A directed graph $G=(V,E)$,
Starting vertex s ,
A non-negative length l_e
for each $e \in E$

Output: $\text{dist}(s, v)$ for every
 $v \in V$

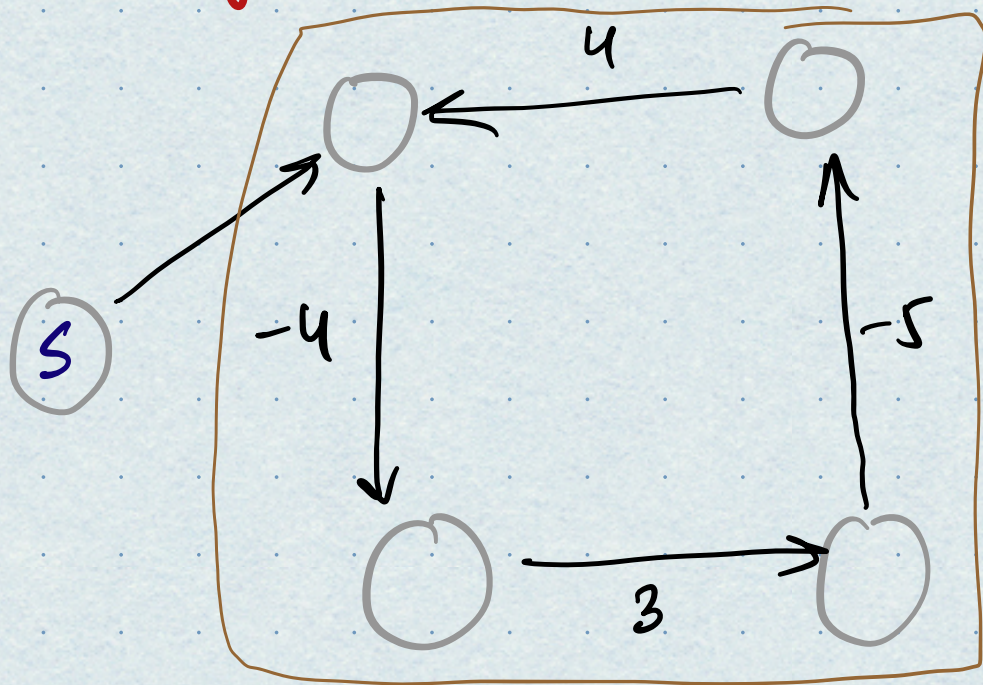
length of the shortest path
from s to v $\begin{cases} \text{if } s \rightsquigarrow v \text{ exists} \\ \infty \text{ otherwise} \end{cases}$

Q:- What about -ve length
cycles?

Dijkstra Fails!

Q. What is the fix?

How to define Shortest paths in the presence of negative cycles



Cycle with weight -2 .

If we include cycles in Shortest paths then length of the Shortest path is undefined (or $-\infty$)

* Then it could go beyond polynomial in terms of run time.

"one can encode longest path" problem.

Idea:- Allow negative length edges but not negative cycles.



But how to quickly check this?

Negative Cycles

Suppose the input graph has no negative cycles.

Then, for any vertex v ,
the shortest $s \rightsquigarrow v$ path
has ? edges?

$$\leq \underline{n-1}$$

Why?

If $\geq n$ edges, then
a cycle exists.

Remove the (non-negative) cycle
 $\rightarrow$ makes the path shorter.

Hence,

No negative cycles $\Rightarrow$

Shortest $s \rightsquigarrow v$ path
has at most $n-1$
edges.

SSSP

I/P :- A directed graph $G=(V,E)$
with start vertex s ,
a possibly negative length
 l_e , for each edge $e \in E$.

Output -

① for all destinations $v \in V$,
compute the length of the
shortest $s \rightsquigarrow v$ path, or

② Output a negative
cycle.

/

This will be the
Excuse for failing
to find a shortest
path.

Optimal Substructure

No inherent sequentiality
in the input (general graph)

But there can be structure - i.e., path
(sequentiality)

Intuition -

Subpath of Shortest
path should itself be
Shortest.

Difficulty

Not fully clear in what order to process the vertices.

How to define "smaller" or "larger" subproblems.?

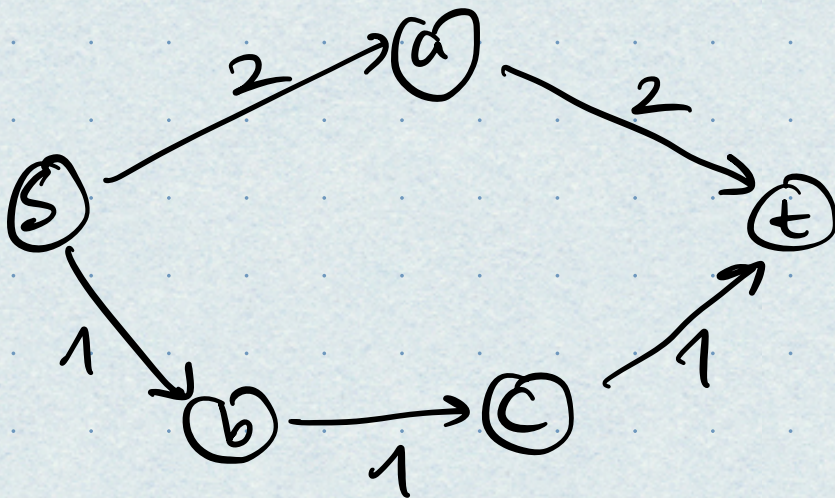


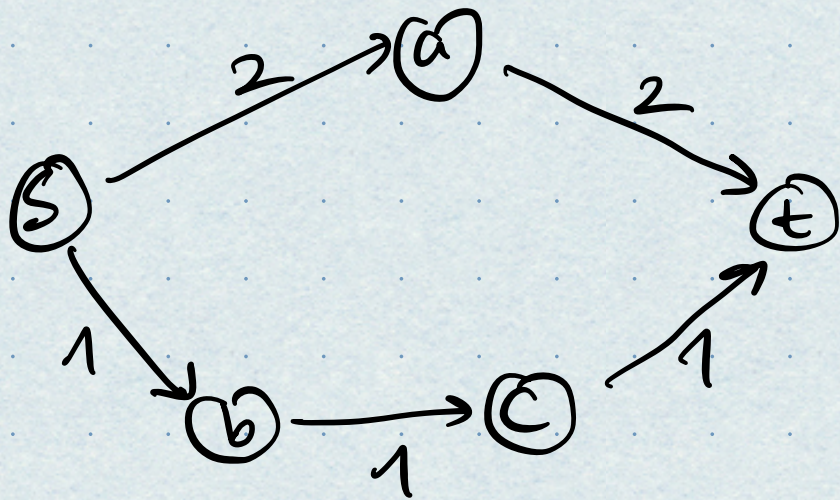
possible that $\text{length of } s \sim w > \text{length } s \sim v$
 $\Downarrow$

of edges in $S \rightsquigarrow w$
 $\leq$ # edges in
 $S \rightsquigarrow v$

Key Idea

Introduce an artificial
parameter : # of edges in
a path.





One
Subprob
for each
destination
and edge budget

$s \rightsquigarrow t$ path

edges = 2

$\Rightarrow s \rightsquigarrow a \rightsquigarrow t$

edges = 3

$\Rightarrow s \rightsquigarrow b \rightsquigarrow c \rightsquigarrow t$

Lemma :- Let $G = (V, E)$
be a directed graph with
edge lengths l_e and start vertex S .
may or may not
have -ve cycles.

For every destination $v \in V$ &
every $i \in \{1, 2, \dots\}$, let P
be the shortest $S \rightsquigarrow v$ path
with at most i edges.

Allow P to
include cycles.

edge budget is finite $\Rightarrow$
of cycles is finite

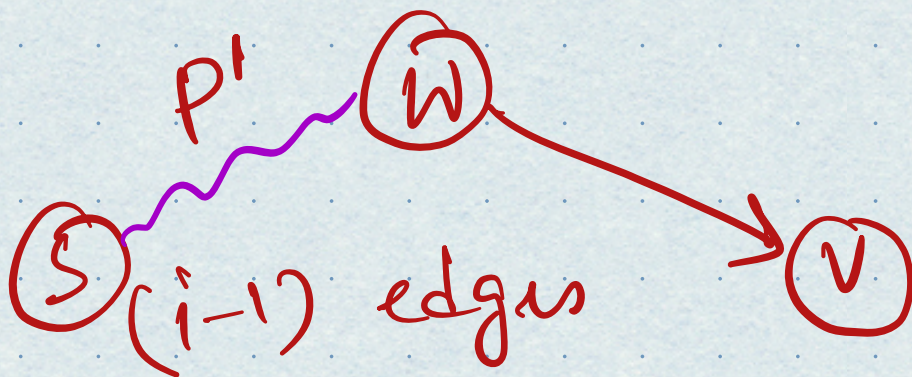
Case -1

p has $(i-1)$ or fewer edges.

$\Rightarrow p$ is the Shortest $s \rightsquigarrow v$ path with at most $(i-1)$ edges.

Case -2

p has i edges.
 $\Rightarrow$



$\Rightarrow$ p^l is the shortest
 $S \rightsquigarrow W$ path with
at most $(i-1)$ edges.

For any given destination v , how many candidates are there for an optimal solⁿ to a subproblem involving v ?

$1 + \text{in-degree}(v)$

inherited from Case 1

from Case 2
for each w
s.t. $(v, w) \in E$

The Recurrence

(cycles allowed)
Define $L_{i,v}$ = length of

shortest $s \rightsquigarrow v$ path
with $\leq i$ edges

with cycles allowed

Defined as $+\infty$ if
no $s \rightsquigarrow v$ path with $\leq i$
edges.

For every $v \in V$ and
every $i \in \{0, 1, 2, \dots\}$

$$L_{i,v} = \min \begin{cases} L_{i-1,v} & \text{// Case-1} \\ \min_{(w,v) \in E} L_{i-1,w} + L_{w,v} & \text{// Case-2} \end{cases}$$

Correctness —

By the optimal
substructure property
(lemma)

If No -ve Cycle

Recall - If G has no negative cycle

shortest path length $\leq (n-1)$ edges

$\Rightarrow$ edge budget $i \leq n-1$

Subproblems

compute $L_{i,v}$, $\forall i \in \{0, \dots, n-1\}$
and all $v \in V$.

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Note

$O(n^2)$ Subproblems
Overall.

$O(n)$ Subproblems per
destination

BELLMAN - FORD

$A = n \times n$ dimensional

array

// Budget

X destination

// base cases

$$A[0, s] = 0,$$

$$A[0, v] = +\infty,$$

$$\forall v \neq s$$

// recurrences

for $i \in \{1, 2, \dots, n-1\}$

for each $v \in V$

$$A[i, v] = \min \begin{cases} A[i-1, v] \\ \min_{\substack{(w, v) \\ \in E}} A[i-1, w] + l_{w, v} \end{cases}$$

$\hat{i} = 0$

	s	v_1	v_2	v_3	v_4
0	0				
1					
2					
3					
4					

$\hat{i} = 0$

	s	v_1	v_2	v_3	v_4
0	0	$+\infty$	$+\infty$	$+\infty$	$+\infty$
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$i=0$

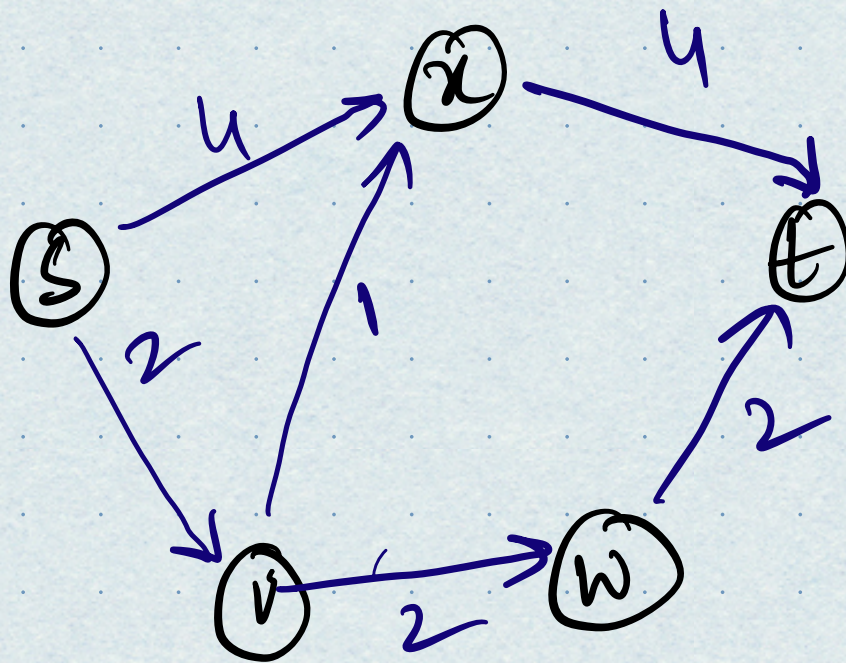
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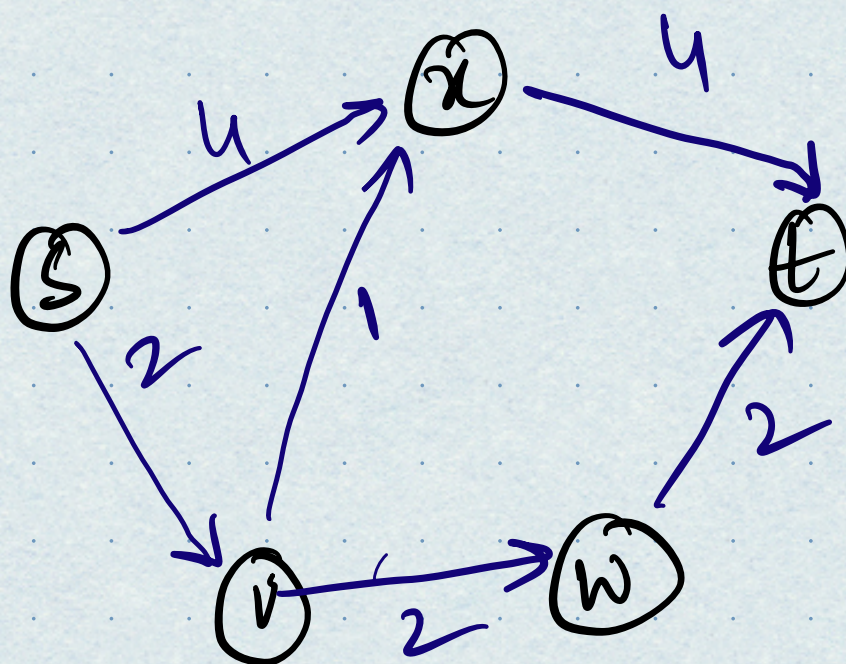
Order is important

$$A[i, v] = \min \left\{ \begin{array}{l} A[i-1, v] \\ \min_{\substack{(w,v) \in E}} A[i-1, w] + l_{w,v} \end{array} \right.$$

If G has no -ve cycle then ALG is Correct.

Ex.



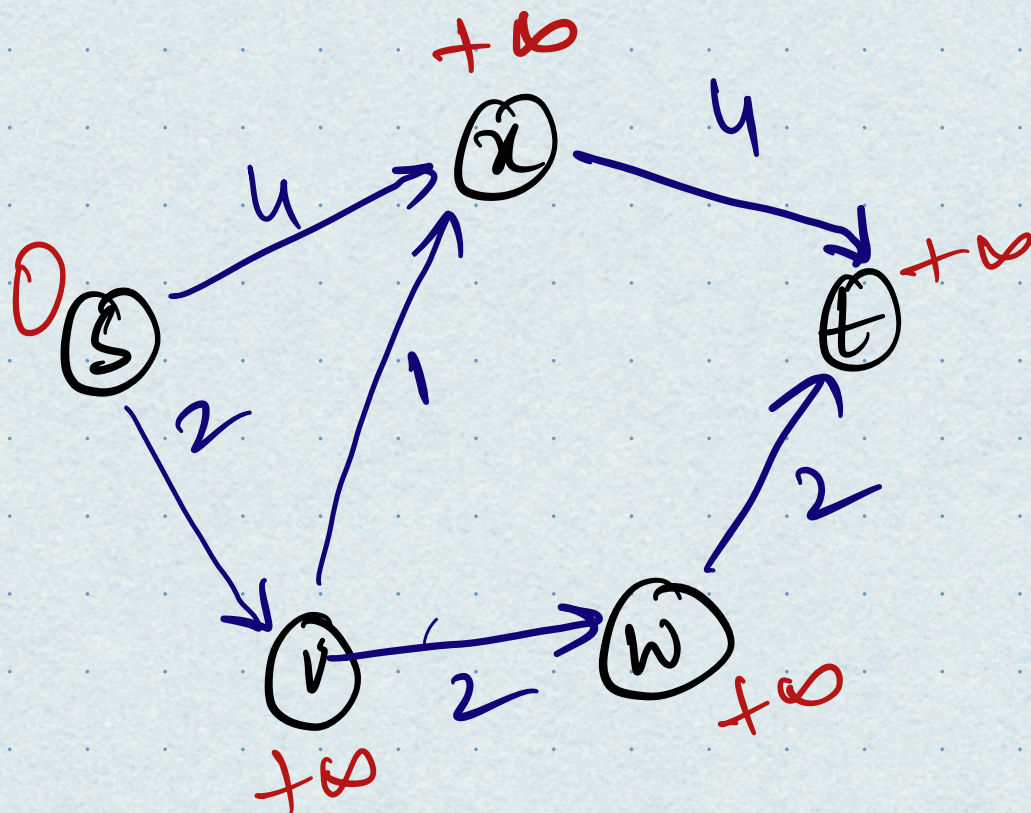


$$\underline{\hat{I} = 0}$$

base case

$$A[0, s] = 0,$$

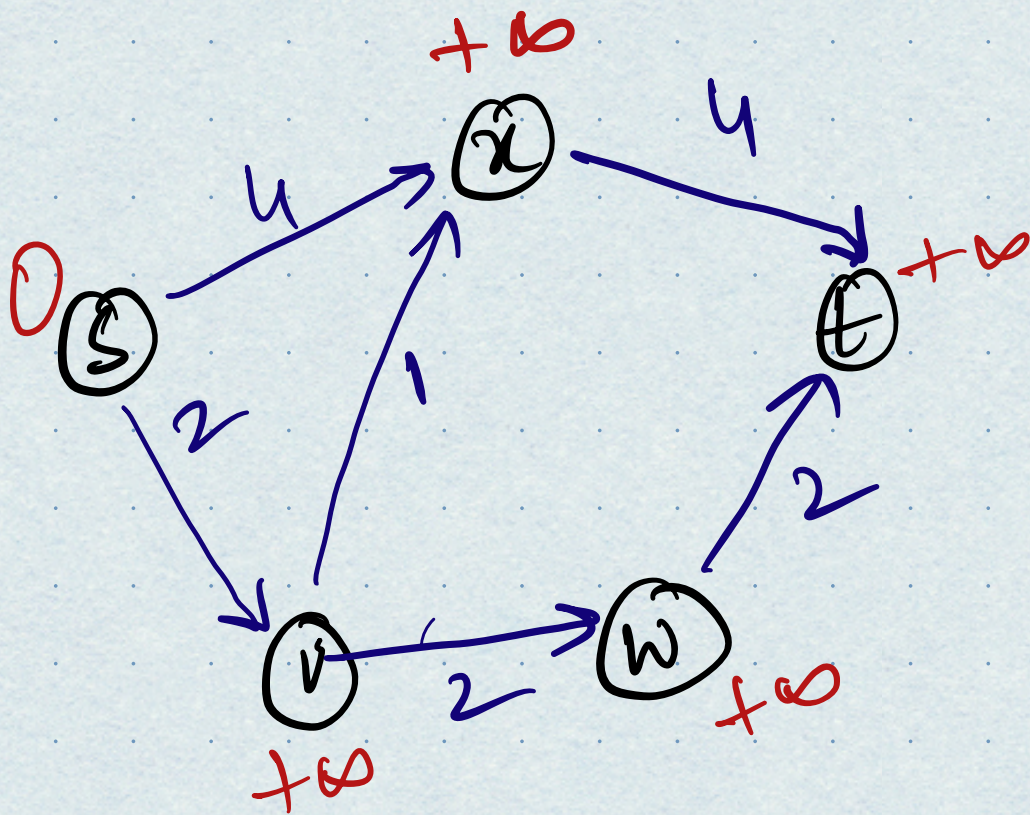
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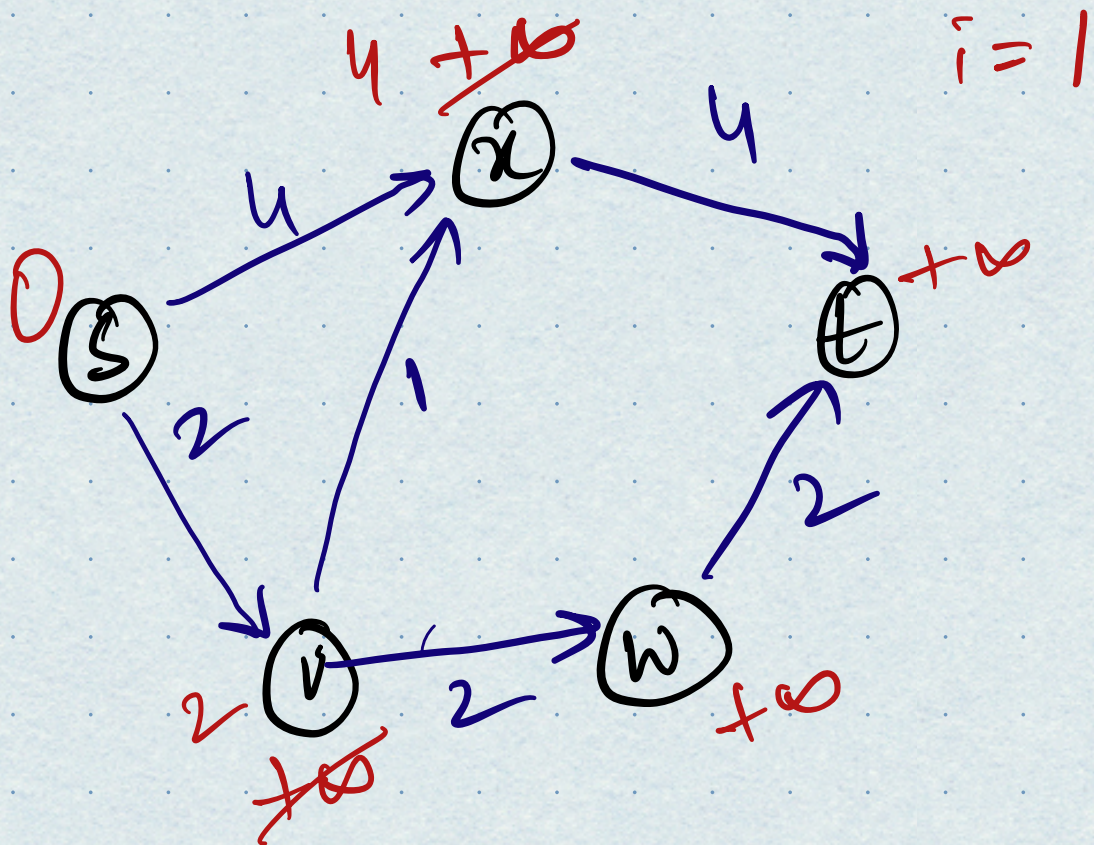
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// recurrences

for $i \in \{1, 2, \dots, n-1\}$
for each $v \in V$

$$A[i, v] = \begin{cases} A[i-1, v] \\ \min_{\substack{(w,v) \in E}} \{ A[i-1, w] + l_{w,v} \} \end{cases}$$

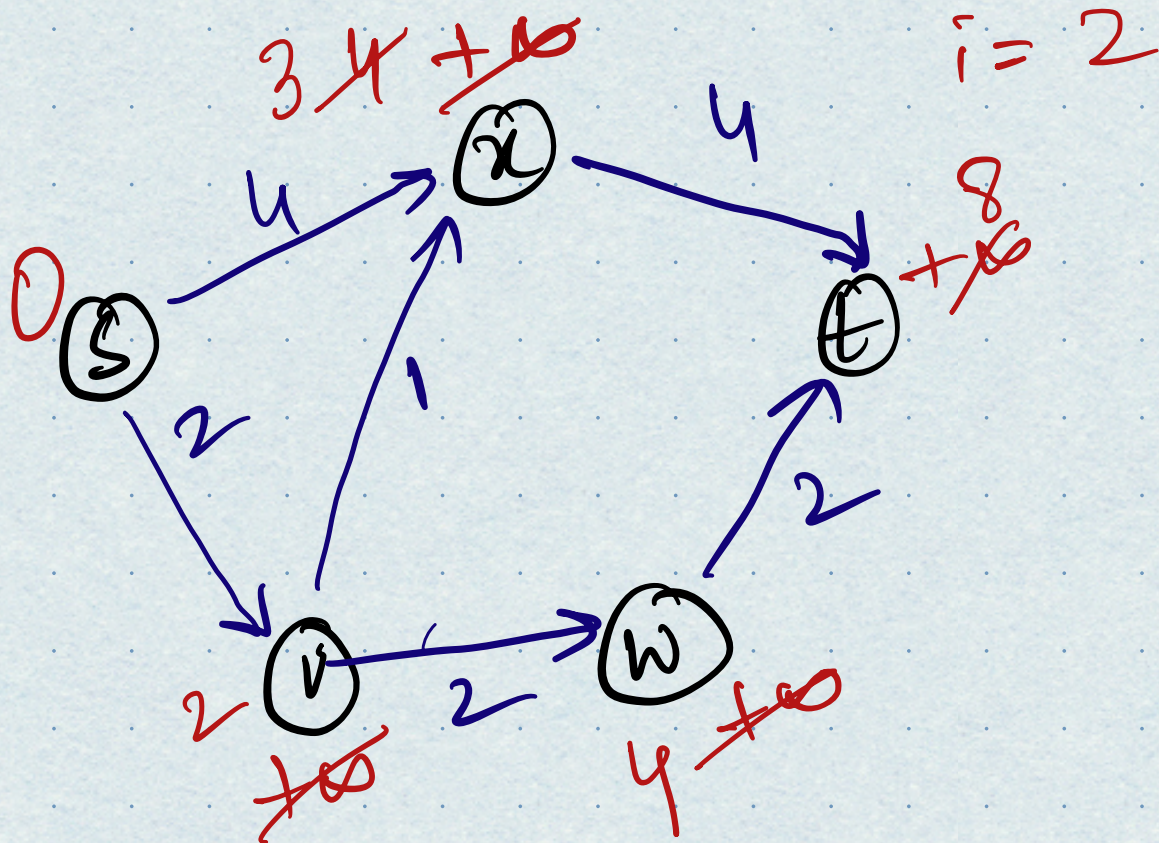


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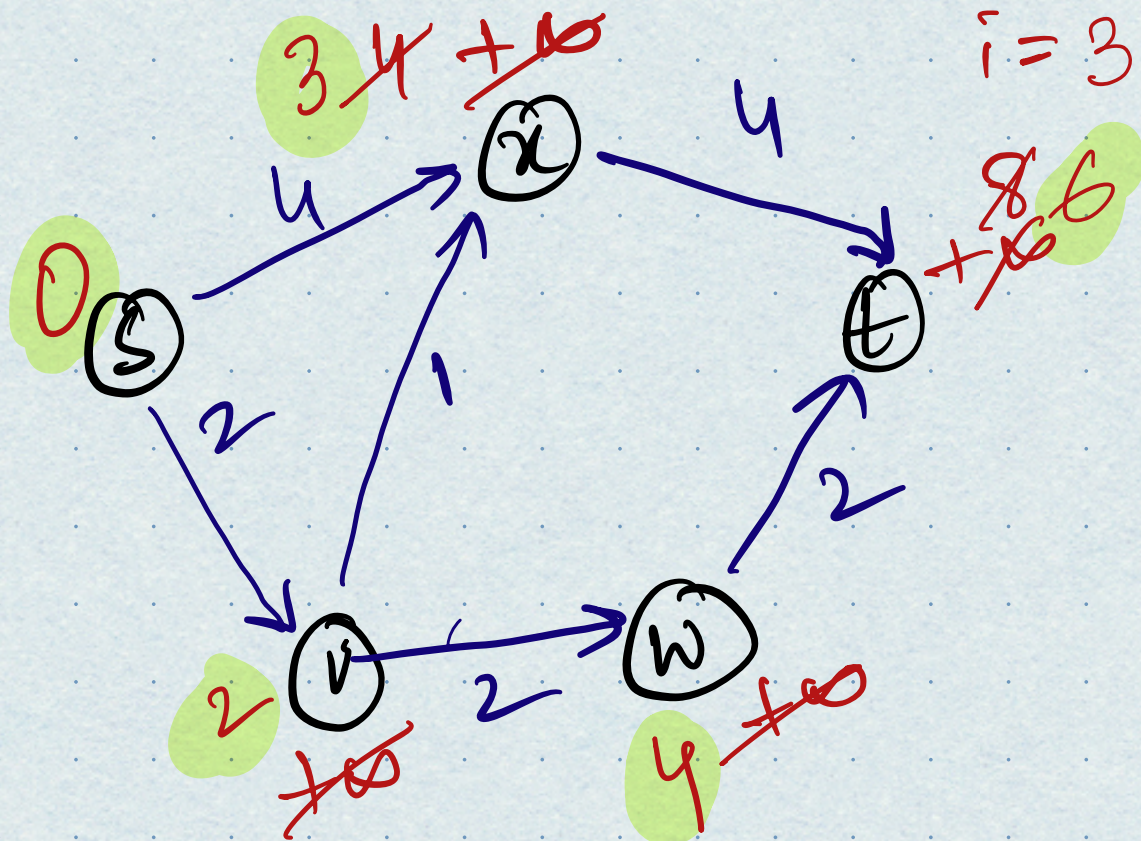


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BELLMAN-FORD

$A = n \times n$ dimensional array

Base Case

$$A[0, \cdot]$$

Recurrence

for $i \in \{1, \dots, n-1\}$

for each $v \in V$

$$A[i, v],$$

Running time?

Running time

$$\underline{O(mn)}$$

$$O\left(n \left(1 + \sum_v \text{in-deg.}(v)\right)\right).$$

$$= O(n^2 + mn) = O(mn)$$

// recurrences

for $i \in \{1, 2, \dots, n-1\}$

for each $v \in V$

$$A[i, v] = \min \begin{cases} A[i-1, v] \\ \min_{\substack{(w, v) \in E}} A[i-1, w] + l_{w, v} \end{cases}$$

What about the
-ve cycles

Claim: - G has no -ve cycle
if and only if in the
Extended Bellman-Ford

algorithm

$$A[n-1, v] = A[n, v], \forall v \in V$$

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How to extend?

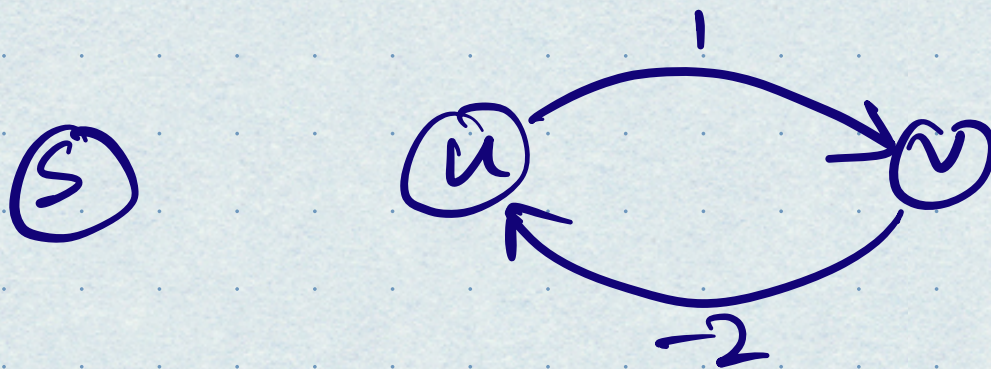
Run the algorithm
for one more iteration
 $i \in \{1, 2, \dots, n-1, n\}$.

Consequence -

Can check existence of a negative cycle by running Bellman-Ford for an extra iteration.

$\Rightarrow$ Run time still $O(m \times n)$.

What if the cycle is not reachable from s ?



claim doesn't hold
anymore.

Fix - Add auxiliary
vertex to detect
-ve cycle, reachable
from s or otherwise.

Claim: - G has no -ve cycle
if and only if in the
Extended Bellman-Ford
algorithm

$$A[n-1, v] = A[n, v], \forall v \in V$$

Proof

$(\Rightarrow)$ If no negative
cycle reachable from
 s , then claim follows
by correctness of
standard Bellman-Ford.

($\Leftarrow$) Assume there is
a path from s to
every other vertex.

[Ex. Try to prove with-
out such assumption]

Recall,

$$d(v) = \min \left\{ A[n-1, v], \min_{(w,v) \in E} d(w) + l_{w,v} \right\}$$

$$\Rightarrow d(v) \leq d(w) + l_{w,v} ?$$
$$\forall (w,v) \in E.$$

Let $d(v)$ - be the common finite value of $A[n-1, v]$ and $A[n, v]$.

$$\Rightarrow d(v) - d(w) \leq l_{w,v} \\ \forall (w, v) \in E.$$

Consider an arbitrary cycle

$$\sum_{(w,v) \in C} l_{w,v} \geq \sum_{(w,v) \in C} d(v) - d(w)$$

$$\begin{matrix} \parallel \\ 0 \end{matrix}$$

